

Descriptive Analysis Report

ON

“Flipkart Sales Analysis”

*Submitted in partial fulfillment of
Bachelor of Computer Application (DS&AI)*



Session: 2026-2027

SCHOOL OF COMPUTER APPLICATION

Babu Banarasi Das University (BBDU)

Submitted To:

Miss. MONIKA MA'AM

(Descriptive Analytics Faculty)

Submitted By:

NAME: ADNAN AHMAD

ROLL NO: 1250258048

BATCH: BCA DS&AI 21

DATE: 15/August/2026

Descriptive Analytics Report: Flipkart Sales Performance

Tracking Best-Sellers, Sales Trends, Ratings, Regional Demand & Discount/Margin Exposure

Note on data: This report is built directly from your uploaded project dataset (*flipkart_sales_enriched.csv*) — **100 orders** across **24 products** in **5 categories**, spanning **4 regions** and **89 distinct order dates** from 12 Feb 2024 to 09 Feb 2025. Unlike a high-volume production feed, this is an order-level sample, so a few sections below are reframed (noted inline) to fit what the data actually contains rather than forcing a fit to the original template.

1. Objective

Build a descriptive analytics layer over the Flipkart order data to answer twenty-six recurring business questions, grouped into the twelve core themes below:

- 1 What are the best-selling products, and how concentrated is demand among them?
- 2 How are total sales trending over the order history?
- 3 How are products rated, and where is rating quality weakest?
- 4 Where is demand concentrated geographically?
- 5 Which products carry the greatest discount and margin exposure?
- 6 Which categories contribute the most to overall revenue and units sold?
- 7 How does customer segment (Online, Retail, Wholesale) influence order volume and order value?
- 8 Which payment methods are most commonly used, and does that vary by region or segment?
- 9 Is there a relationship between unit price and quantity sold across categories?
- 10 Are there seasonal or month-over-month patterns in order volume and revenue?
- 11 Does discount depth correlate with quantity sold or with customer rating?
- 12 Which categories generate the highest total profit under the dataset's flat margin assumption?

Each of these themes is expanded below into its own section, cross-tabulated where useful against region, category, customer segment, and payment method so the same question can be answered from more than one angle rather than a single flat number.

2. How This Would Work at Real Scale

Processing Flipkart-style purchase events at real marketplace volume is an engineering problem before it's an analytics one. A realistic architecture:

Layer	Purpose	Typical tools
Ingestion	Capture each order event in real time	Kafka / Kinesis streams from checkout service
Stream processing	Roll up raw events into per-product, per-region, per-minute aggregates	Spark Structured Streaming / Flink
Storage	Hold both raw events (for audits) and pre-aggregated tables (for speed)	Data lake (S3) + warehouse (Redshift/Snowflake/BigQuery)
Batch aggregation	Nightly rollups: daily sales, rating averages, discount/margin reconciliation	Airflow-orchestrated SQL/dbt jobs
Serving/BI layer	Dashboards refreshed hourly or daily	Looker / Tableau / QuickSight

The report below reflects the *output* of that kind of pipeline — aggregated tables built from the uploaded order data — not raw event-level rows, since a dashboard doesn't render individual transactions usefully.

3. Data Dictionary & Methodology

Every metric in this report traces back to one of the following columns in *flipkart_sales_enriched.csv*. Listing them here keeps every downstream number auditable:

Column	What it captures
Order ID	Unique identifier per order
Product Name / Category	What was purchased, and its category grouping
Price (INR)	Unit price at time of sale
Quantity Sold	Units purchased in that order
Total Sales (INR)	Order revenue (Price × Quantity)
Order Date	Date the order was placed
Payment Method	How the order was paid for (6 methods observed)
Customer Rating	Rating attached to that order, 1-5 scale
Profit (INR)	Profit recorded per order (flat 20% of revenue in this file)

Column	What it captures
Discount %	Discount applied to that order
Customer Segment	Online / Retail / Wholesale buyer classification
Region	North / South / East / West

Methodology note: all figures below are computed directly from these 100 rows with pandas groupby/aggregation — no imputation, extrapolation, or synthetic rows were added. Percentages and correlations are reported to 1-2 decimal places and should be read as descriptive of this specific sample, not as statistically significant findings at production scale.

4. Sales Trend

Across the 89 order dates in the file (Feb 2024 to Feb 2025): **321 units sold**, generating **Rs. 7,768,157 in revenue** (Rs. 77.7 lakh), with order volume fluctuating month to month rather than following a clean daily cycle — a natural consequence of this being a 100-row order sample rather than a full daily transaction feed.

KPI definitions used:

- *Units sold* = sum of Quantity Sold across all orders in the file
- *Revenue* = sum of Total Sales (INR) across all orders
- *Profit* = sum of Profit (INR) as recorded per order, net of discount already applied at time of sale

Quarterly rollup:

Quarter	Units sold	Revenue	Orders
2024Q1	48	Rs. 1,167,757	14
2024Q2	90	Rs. 2,303,729	27
2024Q3	80	Rs. 1,915,262	24
2024Q4	76	Rs. 1,702,838	26
2025Q1	27	Rs. 678,572	9

2024 Q2 and Q3 were the strongest quarters by revenue; 2025 Q1 is a partial quarter in this extract (data ends 09 Feb 2025), so it isn't directly comparable to the full quarters above.

5. Best-Selling Products

Product	Units sold	Revenue	Avg. rating
Jeans	29	Rs. 817,628	3.0
Lipstick	25	Rs. 781,075	2.6
Comic Book	23	Rs. 330,491	3.3
Cookbook	19	Rs. 414,723	3.1
Bedsheet	18	Rs. 342,500	3.4
Perfume	17	Rs. 636,750	2.7
Educational Book	17	Rs. 532,879	2.3
Headphones	16	Rs. 345,019	2.1
Smartwatch	15	Rs. 398,158	3.2
T-Shirt	13	Rs. 267,939	3.6

Jeans led with **29 units** across the sample — noticeably ahead of the next-highest product, which mirrors the "hero SKU" pattern common in marketplaces (a small number of listings drive a disproportionate share of volume).

Ranking logic: units sold, not revenue — a lower-priced item outselling a premium item in volume matters differently to warehousing than it does to finance, so both cuts (units and revenue) are shown as separate leaderboard views above rather than one blended score.

Note: with only 1 order(s) on the most recent date (09 Feb 2025), a "best-sellers today" view (as in a live daily dashboard) isn't statistically meaningful here — the table above uses the full order history instead.

6. Rating Distribution

Average rating across all orders: **2.92 / 5**, with a wide spread (min 1.0, max 4.9) rather than the tight clustering you'd expect from a healthy catalog — **64 of 100 orders (64%)** carry a rating below 3.5. In a real system, this view is most useful **segmented**, not aggregated:

- Products trending *downward* in rolling rating average — the earliest signal of a quality or fulfillment issue
- Products with **high sales volume but sub-3.5 ratings** — the highest-risk combination, since volume amplifies the damage of poor reviews

Product	Units sold	Avg. rating
Jeans	29	3.0

Product	Units sold	Avg. rating
Lipstick	25	2.6
Comic Book	23	3.3
Cookbook	19	3.1
Bedsheet	18	3.4

Rating buckets (all orders):

Rating range	Orders
1.0 - 1.9	20
2.0 - 2.9	28
3.0 - 3.9	32
4.0 - 4.9	20

Ratings are fairly evenly spread across all four bands rather than clustering near the top — 20 orders sit in the lowest band (1.0-1.9), which is as large as the top band (4.0-4.9), pointing to a genuine quality spread rather than a few outliers dragging the average down.

7. Top & Bottom Rated Products

Restricted to products with at least 2 orders, to avoid a single rating standing in for a whole product:

Top rated:

Product	Avg. rating	Units sold	Orders
Sofa Cover	4.2	8	2
Jacket	4.0	10	3
Fiction Novel	3.9	6	2
T-Shirt	3.6	13	4
Laptop	3.5	7	3

Bottom rated:

Product	Avg. rating	Units sold	Orders
Mixer Grinder	1.7	12	4
Hair Dryer	2.0	13	3
Sneakers	2.0	9	3
Headphones	2.1	16	5
Shampoo	2.2	8	3

Mixer Grinder and **Hair Dryer** anchor the bottom of the ratings table at under 2.0 while still selling a meaningful number of units — exactly the high-volume/low-rating combination flagged as highest-risk in Section 6.

8. Regional Demand

Region	Units sold	Revenue	Orders
South	106	Rs. 2,586,177	32
East	84	Rs. 1,780,404	26
West	68	Rs. 1,729,817	22
North	63	Rs. 1,671,759	20

Demand is led by **South** (~106 units) and **East** (~84 units), with the remaining regions long-tailed behind them. This view is what drives:

- Regional warehouse/fulfillment center allocation
- Localized pricing and promotion decisions
- Shipping SLA risk (demand concentrated in fewer regions raises exposure if one region underperforms)

9. Category Performance by Region

Units sold, category × region (rows sum to each category's regional split):

Category	North	South	East	West
Beauty	9	28	27	11
Books	8	28	24	9

Category	North	South	East	West
Clothing	22	11	11	17
Electronics	7	18	10	20
Home & Kitchen	17	21	12	11

Clothing and Electronics skew West/North, while Beauty and Books skew East/South — the kind of regional product-mix signal that would inform which categories to stock more heavily in which fulfillment center.

10. Discount & Margin Exposure

The original report template's "Inventory Status" section tracked stock-out risk — this dataset doesn't include a stock/inventory field, so this section is reframed around the closest available risk signal: discounting.

Average discount applied across all orders: **18.4%**. Products were ranked by average discount depth to flag where margin is being given up most aggressively:

Product	Avg. discount	Total profit	Margin %
Self-Help Book	25.0%	Rs. 15,064	20.0%
T-Shirt	24.2%	Rs. 53,588	20.0%
Cookbook	24.0%	Rs. 82,945	20.0%
Laptop	21.7%	Rs. 52,778	20.0%
Fiction Novel	21.5%	Rs. 53,524	20.0%

Lowest-discount products (least margin given up per order):

Product	Avg. discount	Total profit
Sofa Cover	11.0%	Rs. 63,934
Cookware Set	11.6%	Rs. 31,100
Perfume	14.2%	Rs. 127,350
Bedsheet	15.2%	Rs. 68,500
Smartphone	15.5%	Rs. 12,428

Profit margin holds at a flat 20% across every product in this file (profit appears computed as a fixed share of revenue rather than varying with cost or discount), so discount depth is the more informative signal here — the most actionable version of this view in production would replace the fixed-margin assumption with real per-SKU cost data.

11. Discount Tier Analysis

Orders bucketed by discount depth, with average units and rating per bucket:

Discount tier	Orders	Avg. units/order	Avg. rating
0% - 9%	16	3.0	2.5
10% - 19%	39	3.5	2.9
20% - 29%	39	3.1	3.2
30%+	6	3.2	2.6

No tier stands out as clearly better or worse on either units or rating — consistent with the near-zero correlations reported in Section 20, deeper discounting isn't buying more volume or better reviews in this sample.

12. Category Performance

Category	Units sold	Revenue	% of revenue
Beauty	75	Rs. 2,057,781	26.5%
Books	69	Rs. 1,621,033	20.9%
Clothing	61	Rs. 1,526,386	19.6%
Electronics	55	Rs. 1,340,583	17.3%
Home & Kitchen	61	Rs. 1,222,374	15.7%

Beauty is the leading category, contributing **26% of total revenue** despite the five categories being fairly close in size — no single category dominates the way a hero SKU dominates at the product level.

13. Category Performance by Customer Segment

Units sold, category × customer segment:

Category	Online	Retail	Wholesale
Beauty	28	27	20
Books	19	25	25
Clothing	17	19	25
Electronics	23	7	25
Home & Kitchen	25	15	21

Electronics leans heavily on Online and Wholesale buyers with comparatively little Retail volume, while Books and Beauty are more evenly spread across all three segments.

14. Customer Segment

Segment	Units sold	Revenue	Orders	Avg. order value
Wholesale	116	Rs. 3,126,380	38	Rs. 82,273
Retail	93	Rs. 2,370,225	29	Rs. 81,732
Online	112	Rs. 2,271,553	33	Rs. 68,835

Wholesale leads on revenue, but average order value is close across all three segments — the gap between segments here is driven more by order count than by how much any single order is worth.

15. Customer Segment by Payment Method

Order counts, customer segment × payment method:

Segment	COD	Credit Card	Debit Card	Net Banking	UPI	Wallet
Online	5	8	9	1	6	4
Retail	4	7	1	6	4	7
Wholesale	8	6	5	7	6	6

Wholesale buyers show the most even spread across payment methods, while Online buyers lean slightly more toward Debit Card and Credit Card than the other segments.

16. Payment Method

Payment method	Orders
Credit Card	21
Wallet	17
COD	17
UPI	16
Debit Card	15
Net Banking	14

Credit Card is used most often, but usage is fairly evenly spread across all six methods — no single payment method accounts for more than a fifth of orders, and the mix doesn't shift sharply by region or customer segment in this sample.

By region (order counts):

Region	COD	Credit Card	Debit Card	Net Banking	UPI	Wallet
East	5	5	3	3	4	6
North	3	4	4	2	2	5
South	6	7	4	6	4	5
West	3	5	4	3	6	1

17. Price vs. Quantity Sold

Overall correlation between unit price and quantity sold: **-0.02** — essentially no linear relationship, so higher-priced items in this sample don't systematically sell in smaller quantities (or vice versa). Average unit price by category:

Category	Avg. unit price
Beauty	Rs. 26,402
Clothing	Rs. 25,649
Electronics	Rs. 24,931
Books	Rs. 24,903
Home & Kitchen	Rs. 19,885

By price tier:

Price tier	Orders	Avg. units/order	Avg. rating
Under Rs. 15k	31	3.1	2.6
Rs. 15k - 25k	19	3.6	2.7
Rs. 25k - 35k	24	3.0	3.2
Rs. 35k+	26	3.2	3.2

Ratings trend slightly higher in the two upper price tiers than the two lower ones, though the difference is modest and the sample per tier is small.

18. Order Value Distribution

Orders bucketed by total order value (Price × Quantity):

Order value	Orders	% of orders	Revenue
Under Rs. 25k	22	22%	Rs. 239,949
Rs. 25k - 50k	21	21%	Rs. 845,738
Rs. 50k - 100k	24	24%	Rs. 1,764,095
Rs. 100k - 150k	20	20%	Rs. 2,467,340
Rs. 150k+	13	13%	Rs. 2,451,036

The **Rs. 100k - 150k** band contributes the most revenue despite not having the most orders — a small number of high-value orders carry a disproportionate share of total revenue, the order-level equivalent of the hero-SKU pattern seen in Section 5.

19. Seasonality

Month	Units sold	Revenue	Orders
2024-02	21	Rs. 471,384	6
2024-03	27	Rs. 696,373	8
2024-04	35	Rs. 1,107,705	10
2024-05	29	Rs. 599,344	8
2024-06	26	Rs. 596,680	9

Month	Units sold	Revenue	Orders
2024-07	40	Rs. 894,978	12
2024-08	20	Rs. 517,739	5
2024-09	20	Rs. 502,545	7
2024-10	23	Rs. 691,720	9
2024-11	29	Rs. 544,517	9
2024-12	24	Rs. 466,601	8
2025-01	19	Rs. 379,728	6
2025-02	8	Rs. 298,844	3

Volume peaked in **2024-07** (40 units) and was lowest in **2025-02** (8 units). Month to month movement looks more like normal fluctuation in a small sample than a clean seasonal cycle — a larger extract would be needed to confirm a true seasonal pattern.

20. Discount vs. Quantity and Rating

Correlation between discount % and quantity sold: **0.01** — effectively no relationship, meaning deeper discounts in this sample don't translate into higher unit sales. Correlation between discount % and customer rating: **0.13** — also negligible, so discount depth isn't a meaningful predictor of how an order gets rated either.

21. Category Profit

Category	Total profit	Revenue	Margin %
Beauty	Rs. 411,556	Rs. 2,057,781	20.0%
Books	Rs. 324,207	Rs. 1,621,033	20.0%
Clothing	Rs. 305,277	Rs. 1,526,386	20.0%
Electronics	Rs. 268,117	Rs. 1,340,583	20.0%
Home & Kitchen	Rs. 244,475	Rs. 1,222,374	20.0%

Beauty generates the most total profit (Rs. 411,556), but since every category holds the same flat 20% margin in this file, category profit ranking here just mirrors the revenue ranking in Section 12 rather than reflecting a true cost-driven difference.

22. Full Product Catalog

All 24 products in the file, ranked by revenue — the complete data behind every leaderboard above:

Product	Category	Units	Revenue	Profit	Rating	Discount	Orders
Jeans	Clothing	29	Rs. 817,628	Rs. 163,526	3.0	18.3%	7
Lipstick	Beauty	25	Rs. 781,075	Rs. 156,215	2.6	19.7%	6
Perfume	Beauty	17	Rs. 636,750	Rs. 127,350	2.7	14.2%	6
Educational Book	Books	17	Rs. 532,879	Rs. 106,576	2.3	16.7%	6
Cookbook	Books	19	Rs. 414,723	Rs. 82,945	3.1	24.0%	5
Smartwatch	Electronics	15	Rs. 398,158	Rs. 79,632	3.2	21.1%	7
Headphones	Electronics	16	Rs. 345,019	Rs. 69,004	2.1	17.2%	5
Bedsheet	Home & Kitchen	18	Rs. 342,500	Rs. 68,500	3.4	15.2%	5
Comic Book	Books	23	Rs. 330,491	Rs. 66,098	3.3	17.4%	7
Face Cream	Beauty	12	Rs. 323,278	Rs. 64,656	3.2	20.3%	3
Sofa Cover	Home & Kitchen	8	Rs. 319,671	Rs. 63,934	4.2	11.0%	2
Tablet	Electronics	13	Rs. 271,373	Rs. 54,275	3.0	17.8%	4
T-Shirt	Clothing	13	Rs. 267,939	Rs. 53,588	3.6	24.2%	4
Fiction Novel	Books	6	Rs. 267,618	Rs. 53,524	3.9	21.5%	2
Laptop	Electronics	7	Rs. 263,892	Rs. 52,778	3.5	21.7%	3
Jacket	Clothing	10	Rs. 254,009	Rs. 50,802	4.0	19.3%	3
Mixer Grinder	Home & Kitchen	12	Rs. 236,452	Rs. 47,290	1.7	15.5%	4
Sneakers	Clothing	9	Rs. 186,809	Rs. 37,362	2.0	20.7%	3
Shampoo	Beauty	8	Rs. 179,796	Rs. 35,959	2.2	20.0%	3
Table Lamp	Home & Kitchen	11	Rs. 168,252	Rs. 33,650	3.1	19.8%	4
Cookware Set	Home & Kitchen	12	Rs. 155,499	Rs. 31,100	2.7	11.6%	5
Hair Dryer	Beauty	13	Rs. 136,882	Rs. 27,376	2.0	20.0%	3
Self-Help Book	Books	4	Rs. 75,322	Rs. 15,064	3.6	25.0%	1
Smartphone	Electronics	4	Rs. 62,141	Rs. 12,428	3.2	15.5%	2

23. Day-of-Week Patterns

Orders grouped by day of the week the order was placed:

Weekday	Orders	Units sold	Revenue
Monday	13	44	Rs. 1,254,247
Tuesday	15	47	Rs. 1,067,500
Wednesday	8	23	Rs. 646,994
Thursday	19	62	Rs. 1,575,193
Friday	15	54	Rs. 1,031,892
Saturday	12	44	Rs. 1,158,852
Sunday	18	47	Rs. 1,033,480

Thursday sees the most orders (19), while **Wednesday** sees the fewest (8). With only 100 orders spread across 7 weekdays, this is a mild signal at best — worth re-checking once more order history is available.

24. Correlation Matrix

Pairwise correlation across the dataset's numeric fields, summarizing every relationship checked individually in earlier sections into one view:

	Price	Qty	Revenue	Rating	Profit	Discount %
Price	1.00	-0.02	0.76	0.20	0.76	-0.08
Qty	-0.02	1.00	0.54	-0.02	0.54	0.01
Revenue	0.76	0.54	1.00	0.09	1.00	-0.13
Rating	0.20	-0.02	0.09	1.00	0.09	0.13
Profit	0.76	0.54	1.00	0.09	1.00	-0.13
Discount %	-0.08	0.01	-0.13	0.13	-0.13	1.00

The only strong relationships in the file are mechanical ones — Revenue and Profit correlate at 1.00 with each other (profit is a fixed share of revenue) and Price correlates at 0.76 with both (higher-priced items naturally generate more revenue per unit). Every other pairing, including all three discount and rating relationships, sits close to zero.

25. Sample Order-Level Records

The earliest 30 orders in the file, shown at row level for traceability back to the source data (the report above works from aggregates of rows like these):

Order ID	Product	Date	Qty	Revenue	Rating	Region
ORD00041	Comic Book	12-Feb-24	2	Rs. 5,312	3.7	South
ORD00074	Jacket	13-Feb-24	3	Rs. 102,072	4.6	West
ORD00046	Smartwatch	15-Feb-24	5	Rs. 167,217	2.4	West
ORD00013	Educational Book	23-Feb-24	5	Rs. 33,354	1.2	South
ORD00040	Cookbook	23-Feb-24	5	Rs. 139,809	4.0	North
ORD00068	Smartwatch	26-Feb-24	1	Rs. 23,621	4.7	South
ORD00010	Tablet	14-Mar-24	2	Rs. 53,753	4.0	North
ORD00024	Bedsheet	19-Mar-24	1	Rs. 32,506	2.8	South
ORD00077	Sofa Cover	22-Mar-24	5	Rs. 231,691	3.4	North
ORD00019	Cookbook	22-Mar-24	5	Rs. 21,993	4.0	South
ORD00071	Face Cream	24-Mar-24	4	Rs. 94,862	2.0	South
ORD00076	Fiction Novel	25-Mar-24	3	Rs. 128,943	3.8	North
ORD00017	Cookbook	30-Mar-24	2	Rs. 41,635	2.3	East
ORD00016	Mixer Grinder	30-Mar-24	5	Rs. 90,989	1.6	South
ORD00091	Bedsheet	01-Apr-24	4	Rs. 137,174	4.4	East
ORD00056	Jeans	04-Apr-24	5	Rs. 195,883	4.1	West
ORD00069	Educational Book	08-Apr-24	3	Rs. 116,814	3.4	East
ORD00088	Cookware Set	12-Apr-24	5	Rs. 17,160	3.4	North
ORD00012	Lipstick	13-Apr-24	5	Rs. 243,904	1.3	South
ORD00031	Perfume	14-Apr-24	1	Rs. 47,651	3.6	East
ORD00045	Shampoo	14-Apr-24	2	Rs. 63,700	2.8	South
ORD00029	Lipstick	16-Apr-24	5	Rs. 106,467	2.2	East
ORD00018	Perfume	22-Apr-24	4	Rs. 176,245	2.5	South
ORD00070	Cookware Set	28-Apr-24	1	Rs. 2,708	1.2	South
ORD00035	Laptop	01-May-24	2	Rs. 98,711	3.5	South

Order ID	Product	Date	Qty	Revenue	Rating	Region
ORD00080	Face Cream	01-May-24	4	Rs. 184,047	3.3	East
ORD00067	Hair Dryer	06-May-24	4	Rs. 16,903	2.5	South
ORD00058	Self-Help Book	16-May-24	4	Rs. 75,322	3.6	South
ORD00003	Face Cream	18-May-24	4	Rs. 44,369	4.4	South
ORD00094	Cookbook	23-May-24	2	Rs. 53,152	2.0	South

26. Regional & Category Leaders

The single best-selling product within each region and each category, by units sold:

By region:

Region	Top product	Units sold
East	Comic Book	16
South	Cookbook	12
North	Jeans	11
West	Jeans	10

By category:

Category	Top product	Units sold
Clothing	Jeans	29
Beauty	Lipstick	25
Books	Comic Book	23
Home & Kitchen	Bedsheet	18
Electronics	Headphones	16

Jeans leads both the Clothing category overall and two of the four regions (North and West) outright — the clearest single-product concentration signal in the whole dataset.

27. Highest & Lowest Value Orders

The 10 highest-value and 10 lowest-value individual orders in the file, by Total Sales (INR):

Highest value:

Order ID	Product	Order value	Qty	Region
ORD00012	Lipstick	Rs. 243,904	5	South
ORD00011	Perfume	Rs. 232,048	5	West
ORD00077	Sofa Cover	Rs. 231,691	5	North
ORD00030	Lipstick	Rs. 196,388	4	North
ORD00056	Jeans	Rs. 195,883	5	West
ORD00023	Educational Book	Rs. 193,033	4	South
ORD00080	Face Cream	Rs. 184,047	4	East
ORD00018	Perfume	Rs. 176,245	4	South
ORD00046	Smartwatch	Rs. 167,217	5	West
ORD00020	T-Shirt	Rs. 162,829	5	South

Lowest value:

Order ID	Product	Order value	Qty	Region
ORD00081	Perfume	Rs. 2,098	1	East
ORD00070	Cookware Set	Rs. 2,708	1	South
ORD00042	Headphones	Rs. 3,862	4	West
ORD00009	Jacket	Rs. 3,975	2	North
ORD00041	Comic Book	Rs. 5,312	2	South
ORD00050	Comic Book	Rs. 5,566	4	East
ORD00075	Sneakers	Rs. 6,168	4	North
ORD00064	Table Lamp	Rs. 6,474	3	East
ORD00015	Comic Book	Rs. 6,839	1	South
ORD00026	T-Shirt	Rs. 8,212	4	West

The highest-value orders aren't simply the highest-quantity ones — several top-10 orders by value carry only 4-5 units, meaning unit price is doing most of the work, not order size. This reinforces the Section 17 finding that price and quantity move independently in this sample.

28. Category by Payment Method

Order counts, category × payment method:

Category	COD	Credit Card	Debit Card	Net Banking	UPI	Wallet
Beauty	4	5	2	3	3	4
Books	3	5	4	3	3	3
Clothing	3	3	3	2	1	5
Electronics	4	5	3	1	5	3
Home & Kitchen	3	3	3	5	4	2

Electronics sees the most UPI usage of any category, while Home & Kitchen leans hardest on Net Banking — both mild patterns given the small per-cell counts, but worth watching as the order history grows.

29. Data Quality & Limitations

Worth keeping in mind when reading the sections above:

- **Sample size:** 100 orders is enough to spot directional patterns but too small for the correlations and tier breakdowns above to be statistically conclusive — treat them as descriptive, not predictive.
 - **Flat profit margin:** Profit (INR) equals exactly 20% of Total Sales (INR) for every single order, which means category- and product-level profit rankings in this report are mathematically identical to revenue rankings. Real per-SKU cost data would be needed for a genuine margin analysis.
 - **Sparse dates:** 100 orders span 89 distinct dates over roughly a year, so daily-level views (as in the original report template this was adapted from) aren't meaningful here; monthly and quarterly rollups are used instead.
 - **No inventory field:** stock levels aren't tracked in this file, so stock-out risk (Section 10 in the original template) is reframed around discount exposure instead.
 - **Small per-bucket counts:** several breakdowns above (price tiers, discount tiers, monthly figures) rest on 10-40 orders per bucket — enough to describe this sample, not enough to generalize with confidence.
 - **Ratings are order-level, not product-level:** each rating in the file is tied to a specific order rather than a running product rating, so a product's "average rating" above is really the average across its handful of orders — products with only 1-2 orders (see Section 22) carry the least reliable averages.
 - **No returns or cancellations field:** Total Sales (INR) and Profit (INR) appear to reflect completed sales only; if returns or cancellations exist upstream, they aren't netted out anywhere in this extract.
-

30. Summary of Key Metrics

Metric	Value
Total orders in file	100
Total units sold	321
Total revenue	Rs. 7,768,157
Total profit	Rs. 1,553,631
Average customer rating	2.92 / 5
Top-selling product	Jeans (29 units)
Top category by revenue	Beauty (26%)
Top region by units	South (106 units)
Top customer segment	Wholesale
Most-used payment method	Credit Card
Categories tracked	5
Regions tracked	4
Orders with rating below 3.5	64 of 100 (64%)
Average discount applied	18.4%
Price ↔ quantity correlation	-0.02
Discount ↔ quantity correlation	0.01
Discount ↔ rating correlation	0.13

31. Next Steps to Move From Descriptive → Predictive

This report answers *what happened* in the data provided. The natural next layer, if useful:

- **Demand forecasting** per product/region to anticipate reorder timing (would need an actual inventory feed)
- **Anomaly detection** on rating drops or unusual discount spikes
- **Cohort/retention analysis** by Customer Segment (Online / Retail / Wholesale), not just order counts
- **True margin analysis** once real per-product cost data replaces the fixed 20% profit assumption seen in this file

- **Statistical significance testing** on the tier and correlation breakdowns above once the sample grows beyond a few hundred orders
- **Rating root-cause analysis** on the lowest-rated products (Mixer Grinder, Hair Dryer) to identify whether the driver is product quality, fulfillment, or something else

Happy to build out any of these next, or refresh this report if you upload a larger or updated extract — the same structure above would apply directly.

32. Glossary of Terms

Term	Meaning
AOV	Average Order Value — mean Total Sales (INR) per order
Correlation	A -1 to +1 measure of linear relationship; near 0 means no linear relationship
Hero SKU	A single product that drives disproportionate volume relative to the rest of the catalog
Margin	Profit as a percentage of revenue
Segment	Customer Segment column: Online, Retail, or Wholesale
Tier / Bucket	A range used to group continuous values (price, discount, order value) for comparison
Cross-tab	A table showing counts or sums broken down by two variables at once, e.g. category × region
Long-tailed	A distribution where a few items account for most of the volume and the rest trail off gradually
Aggregation	Combining many individual order rows into a single summary figure (a sum, average, or count)
Pareto effect	The pattern where a small share of products or categories drives a large share of total volume
Quarter (Q1-Q4)	A three-month calendar grouping used for the rollups in Section 4
SLA	Service Level Agreement — here, a shipping/fulfillment commitment tied to regional demand

About This Report

This report was generated directly from *flipkart_sales_enriched.csv* and reflects the 100 orders present in that file as of the time of generation. Every number above is reproducible by re-running the same aggregations against the source file — nothing here was estimated or carried over from a template. If the underlying data changes (more orders, more dates, a real inventory feed, per-product cost data), every section in this report would need to be recomputed rather than edited by hand, since each table and callout above is a direct function of the current 100-row extract.